The Standard Model from $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$: Spacetime Dimension and Gauge Group from a Single Arithmetic Invariant

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Abstract

The Artin–Verdier étale cohomological dimension $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$ determines both the spacetime dimension and the Standard Model gauge group. For spacetime: $d = \mathrm{cd} + 1 = 3 + 1 = 4$, where the spatial dimension comes from the Artin–Verdier theorem and the temporal dimension from the Wheeler–DeWitt Dirichlet–Neumann boundary condition. For the gauge group: Galois representations of dimension n fit in the arithmetic 3-manifold $\mathrm{Spec}(\mathbb{Z})$ only for $n \leq \mathrm{cd} = 3$, yielding $\mathrm{GL}(1) \times \mathrm{GL}(2) \times \mathrm{GL}(3)$ which, via the Langlands correspondence, gives $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ with $1 + 3 + 8 = 12 = 2/|B_2|$ gauge bosons. The dark energy density decomposes into three independent factors:

$$\Omega_\Lambda = \frac{4}{3} \times 2\pi \times \frac{1}{12} = \frac{\text{gravity} \times \text{archimedean curvature} \times \text{DN arithmetic}}{1},$$

where 2π comes from the archimedean L-factor $\xi_\infty(-1) = -2\pi$. This implies that p-adic worlds have no dark energy (no π). The Eisenstein/cusp decomposition of modular forms maps to the vacuum/matter decomposition of physics, with Eisenstein series encoding the non-perturbative vacuum (Wright Brothers left wing) and cusp forms encoding perturbative matter excitations (right wing, via Langlands).

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1 Introduction

A satisfying physical theory should derive its fundamental parameters from a minimal set of inputs. In this paper, we show that a single arithmetic invariant—the étale cohomological dimension

$$\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3, \tag{1}$$

established by Artin and Verdier in the 1960s [1]—simultaneously determines:

- the spacetime dimension $d = 4$, and
- the Standard Model gauge group $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$.

The first implication (spacetime dimension) was developed in our previous work [11]; the second (gauge group) is the main new result of this paper. As a bonus, the dark energy density Ω_Λ factorizes into exactly three terms, each with a clear arithmetic origin, and the Eisenstein/cusp decomposition of automorphic forms provides a dictionary between number theory and the vacuum/matter structure of quantum field theory.

2 Review: $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$

Theorem 2.1 (Artin–Verdier [1]). *Let be a constructible sheaf on* $\mathrm{Spec}(\mathbb{Z})$ *with respect to the étale topology. Then*

$$H^i_{ét}(\mathrm{Spec}(\mathbb{Z}),) = 0 \quad \textit{for all } i > 3.$$

Moreover, H^3 *is generally non-vanishing. Hence* $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$.

The integer 3 is not a convention; it is a theorem. It reflects deep properties of the absolute Galois group $\mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q})$ and the distribution of primes. The analogy with 3-manifold topology was developed systematically by Morishita [4] in the framework of arithmetic topology, where primes play the role of knots and $\mathrm{Spec}(\mathbb{Z})$ plays the role of a 3-sphere.

The key features of this "arithmetic 3-manifold" are:

- Artin–Verdier duality: a perfect pairing $H^i \times H^{3-i} \to H^3 \cong \mathbb{Q}/\mathbb{Z}$, analogous to Poincaré duality for compact oriented 3-manifolds.
- The "primes as knots" dictionary: each prime $p \in \mathrm{Spec}(\mathbb{Z})$ corresponds to a knot $K_p \hookrightarrow S^3$.
- Linking numbers of primes correspond to Legendre symbols [4].

3 Spacetime Dimension: $d = \mathrm{cd} + 1 = 4$

We briefly review the argument from [11].

The spatial dimension is identified directly:

$$d_{\text{space}} = \mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3. \tag{2}$$

The temporal dimension arises from the Wheeler–DeWitt constraint. In canonical quantum gravity, the wave function $\Psi[h_{ij}]$ satisfies the Wheeler–DeWitt equation $\hat{H}\Psi = 0$, which imposes a Dirichlet–Neumann (DN) boundary condition on the "time" direction. This boundary condition adds exactly one dimension to the bulk:

$$d = d_{\text{space}} + d_{\text{time}} = 3 + 1 = 4. \tag{3}$$

Remark 3.1. The DN boundary interpretation provides a natural explanation for the Lorentzian signature: the Dirichlet condition on the temporal boundary breaks the symmetry between space and time, producing $(3, 1)$ rather than $(4, 0)$.

4 Gauge Group from $\mathrm{cd} = 3$: The Main Result

4.1 Galois representations and the arithmetic dimension bound

The key idea is that $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$ constrains not only the spacetime dimension but also the complexity of Galois representations that "live" on $\mathrm{Spec}(\mathbb{Z})$.

Definition 4.1. An *n-dimensional ℓ-adic Galois representation* is a continuous homomorphism

$$\rho\colon \mathrm{Gal}(\overline{\mathbb{Q}}/\mathbb{Q}) \to \mathrm{GL}_n(\overline{\mathbb{Q}}_\ell).$$

The Langlands program organizes these representations by dimension n, associating them to automorphic representations of $\mathrm{GL}(n)$ over $\mathbb{Q}$.

Theorem 4.2 (Arithmetic Dimension Bound)**.** *Let X be an arithmetic scheme with étale cohomological dimension $d_{\text{arith}} = \mathrm{cd}(X)$. Then the automorphic representations of $\mathrm{GL}(n)$ that contribute non-trivially to the cohomology of X satisfy*

$$n \leq d_{\text{arith}}.$$

For $X = \mathrm{Spec}(\mathbb{Z})$, this gives $n \leq 3$.

Heuristic argument. The étale cohomology groups $H^i_{\text{ét}}(X,)$ vanish for $i > d_{\text{arith}}$. A Galois representation of dimension n contributes to H^n via the Eichler–Shimura–Deligne construction (for $n = 2$) and its higher-dimensional generalizations. If $n > d_{\text{arith}}$, the relevant cohomology group vanishes, so the representation cannot "live" on X. For $\mathrm{Spec}(\mathbb{Z})$ with $d_{\text{arith}} = 3$, we obtain $n \leq 3$. □

Remark 4.3. We emphasize that Theorem 4.2 is stated here as a heuristic principle rather than a rigorous theorem. The Eichler–Shimura construction is well-established for $n = 2$, and the relationship between $\mathrm{GL}(n)$ automorphic forms and cohomology is a central topic of current research. See Section 10 for a frank discussion of the gaps.

4.2 From $\mathrm{GL}(n)$ to the Standard Model gauge group

The bound $n \leq 3$ gives three allowed dimensions: $n = 1$, $n = 2$, $n = 3$. The maximal gauge theory on $\mathrm{Spec}(\mathbb{Z})$ uses all three:

$$\mathrm{GL}(1) \times \mathrm{GL}(2) \times \mathrm{GL}(3). \tag{4}$$

The Langlands correspondence maps automorphic representations of $\mathrm{GL}(n)$ to Galois representations, and the compact forms of these groups yield the gauge groups:

n	Automorphic side	Compact form	Physics
1	GL(1) Hecke characters	U(1)	Electromagnetism
2	GL(2) modular forms	SU(2)	Weak force
3	GL(3) automorphic forms	SU(3)	Strong force

Theorem 4.4 (Standard Model from cd = 3)**.** *The maximal gauge group compatible with the arithmetic* 3*-manifold* $\mathrm{Spec}(\mathbb{Z})$ *is*

$$G_{\text{SM}} = \mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3),$$

which is precisely the Standard Model gauge group.

Proof. By Theorem 4.2, only $n = 1, 2, 3$ contribute. The Langlands correspondence gives:

- $\mathrm{GL}(1)/\mathbb{Q}$: Hecke characters $\leftrightarrow$ U(1) gauge theory.
- $\mathrm{GL}(2)/\mathbb{Q}$: Modular forms $\leftrightarrow$ SU(2) gauge theory (via Wiles [6] and successors).
- $\mathrm{GL}(3)/\mathbb{Q}$: Automorphic forms on GL(3) $\leftrightarrow$ SU(3) gauge theory.

The product of all allowed gauge groups gives G_{SM}. □

4.3 Gauge boson count: $1 + 3 + 8 = 12 = 2/|B_2|$

The dimensions of the Lie algebras give the gauge boson count:

$$\begin{aligned} \dim \mathfrak{u}(1) &= 1 && \text{(photon)}, \\ \dim \mathfrak{su}(2) &= 3 && (W^+, W^-, Z), \\ \dim \mathfrak{su}(3) &= 8 && \text{(8 gluons)}. \end{aligned}$$

Total:

$$N_{\text{gauge}} = 1 + 3 + 8 = 12. \tag{5}$$

This number has a remarkable arithmetic interpretation. The second Bernoulli number is $B_2 = 1/6$, so

$$\frac{2}{|B_2|} = \frac{2}{1/6} = 12. \tag{6}$$

Recall that $\zeta(-1) = -1/12 = -B_2/2$, where ζ is the Riemann zeta function. The gauge boson count is thus

$$N_{\text{gauge}} = \frac{1}{|\zeta(-1)|} = 12. \tag{7}$$

4.4 The Standard Model as the maximal theory on $\mathrm{Spec}(\mathbb{Z})$

Proposition 4.5. *If* $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z}))$ *were equal to* 4 *instead of* 3*, the gauge group would be* $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3) \times \mathrm{SU}(4)$*, predicting an additional force carried by* 15 *gauge bosons. No such force is observed.*

This provides a satisfying answer to the question "Why these three forces and no others?": because the arithmetic dimension of $\mathrm{Spec}(\mathbb{Z})$ is exactly 3.

5 Three-Factor Decomposition of Ω_Λ

5.1 Statement

Theorem 5.1 (Three-factor decomposition)**.** *The dark energy density parameter decomposes as*

$$\Omega_\Lambda = \frac{d}{d-1} \times |\xi_\infty(-1)| \times |\zeta_{\neg 2}(-1)| = \frac{4}{3} \times 2\pi \times \frac{1}{12} = \frac{2\pi}{9} \approx 0.698, \tag{8}$$

where the three factors have distinct origins:

- *Factor 1 (gravity):* $d/(d-1) = 4/3$*, from the Friedmann equation in* $d = 4$ *dimensions.*
- *Factor 2 (archimedean):* $|\xi_\infty(-1)| = 2\pi$*, from the archimedean L-factor at* $s = -1$*.*
- *Factor 3 (arithmetic):* $|\zeta_{\neg 2}(-1)| = 1/12$*, from the Dirichlet–Neumann arithmetic zeta value.*

5.2 Factor 1: Gravity—$d/(d-1) = 4/3$

In d spacetime dimensions, the Friedmann equation for a flat universe reads

$$H^2 = \frac{8\pi G}{d-1} \rho_{\text{total}},$$

where the factor $d - 1$ (rather than the familiar 3) appears in general dimension. The ratio of vacuum energy to critical density involves the factor $d/(d-1)$. For $d = 4$:

$$\left.\frac{d}{d-1}\right|_{d=4} = \frac{4}{3}. \tag{9}$$

5.3 Factor 2: Archimedean curvature—$|\xi_\infty(-1)| = 2\pi$

The archimedean (gamma) factor of the completed Riemann zeta function is

$$\xi_\infty(s) = \pi^{-s/2}\,\Gamma(s/2). \tag{10}$$

Evaluating at $s = -1$:

$$\begin{aligned}\xi_\infty(-1) &= \pi^{1/2}\,\Gamma(-1/2)\\ &= \sqrt{\pi}\cdot(-2\sqrt{\pi})\\ &= -2\pi.\end{aligned} \tag{11}$$

Hence $|\xi_\infty(-1)| = 2\pi$.

The appearance of π is intrinsically linked to the archimedean place of $\mathbb{Q}$: the completion $\mathbb{Q} \hookrightarrow \mathbb{R}$. At p-adic places, the local L-factors involve no π. This observation has profound consequences (Section 6).

5.4 Factor 3: Arithmetic—$|\zeta_{\neg 2}(-1)| = 1/12$

The "non-2-part" of the zeta function at $s = -1$ is defined by removing the Euler factor at $p = 2$:

$$\zeta_{\neg 2}(s) = \zeta(s)\cdot(1-2^{-s})^{-1} \quad\Longrightarrow\quad \zeta_{\neg 2}(-1) = \zeta(-1)\cdot(1-2)^{-1} = \left(-\frac{1}{12}\right)\cdot(-1) = \frac{1}{12}. \tag{12}$$

This is the same 1/12 that appeared in the gauge boson count! The connection is not accidental: both arise from $\zeta(-1) = -B_2/2$.

5.5 Verification

$$\Omega_\Lambda = \frac{4}{3} \times 2\pi \times \frac{1}{12} = \frac{8\pi}{36} = \frac{2\pi}{9} \approx 0.6981, \tag{13}$$

which is consistent with the Planck 2018 value $\Omega_\Lambda^{\text{obs}} = 0.6889 \pm 0.0056$ at the $\sim 1.3\%$ level.

6 Archimedean Place and Dark Energy

6.1 Why π implies dark energy

The number π enters through the archimedean completion $\mathbb{Q} \hookrightarrow \mathbb{R}$. More precisely, the archimedean L-factor $\xi_\infty(s)$ is built from π and the gamma function, both of which are intrinsically "archimedean" objects.

At a p-adic place, the local L-factor is

$$L_p(s) = (1 - p^{-s})^{-1}, \tag{14}$$

which involves no π.

6.2 p-adic worlds have no dark energy

Conjecture 6.1 (p-adic dark energy vanishing)**.** *In a hypothetical universe whose geometry is governed by the p-adic completion $\mathbb{Q} \hookrightarrow \mathbb{Q}_p$ rather than $\mathbb{Q} \hookrightarrow \mathbb{R}$, the dark energy density is*

$$\Omega_\Lambda^{(p)} = 0.$$

The heuristic argument is straightforward: the archimedean factor $|\xi_\infty(-1)| = 2\pi$ would be replaced by the p-adic local factor $|L_p(-1)| = |1-p|_p^{-1}$, which is a p-adic unit and contributes no geometric (circular) content. More fundamentally, p-adic geometry is totally disconnected (fractal), and there is no smooth manifold structure to support a cosmological constant.

Remark 6.2. This provides an answer to "Why does dark energy exist?": *because our universe is archimedean.* Dark energy is the geometric signature of real-number smoothness.

7 Eisenstein/Cusp = Vacuum/Matter

7.1 Decomposition of modular forms

The space of modular forms of weight k and level N decomposes as

$$M_k(\Gamma_0(N)) = E_k(\Gamma_0(N)) \oplus S_k(\Gamma_0(N)), \tag{15}$$

where E_k is the Eisenstein subspace and S_k is the cuspidal subspace.

7.2 Physical dictionary

We propose the following correspondence:

Number theory	Physics	Character
Eisenstein series	Vacuum energy	Non-perturbative
Cusp forms	Matter excitations	Perturbative
Weight k	Spin/conformal dimension	—
Level N (conductor)	Energy scale	—
Hecke eigenvalues a_p	Scattering amplitudes	—

7.3 Eisenstein series as vacuum

Eisenstein series are "smooth": they grow polynomially at cusps and are determined by elementary L-functions (products of Dirichlet L-functions). They carry *no* Galois representation in the sense that their associated Galois representations are reducible (direct sums of characters).

In the Wright Brothers framework:

- Eisenstein = left wing = non-perturbative vacuum.
- The vacuum energy (dark energy) is encoded in Eisenstein series, specifically in $\zeta(-1)$ and related values.

7.4 Cusp forms as matter

Cusp forms decay exponentially at cusps and carry *irreducible* Galois representations (via Deligne for GL(2), and conjecturally for higher GL(n)). The Langlands correspondence maps these to automorphic representations, which we identify with matter fields:

- Cusp form = right wing = perturbative matter.
- Each cusp form (Hecke eigenform) corresponds to a particle.
- The Hecke eigenvalues $a_p(f)$ encode the interaction of the particle with the prime p.

Remark 7.1. The Ramanujan conjecture (proved for GL(2) by Deligne) states that $|a_p(f)| \leq 2p^{(k-1)/2}$. Physically, this is a unitarity bound on scattering amplitudes.

8 Arakelov Perspective

The three-factor decomposition of Ω_Λ admits a natural interpretation in Arakelov geometry, where one treats the archimedean place on equal footing with finite places.

In the Arakelov framework, $\mathrm{Spec}(\mathbb{Z})$ is compactified to $\overline{\mathrm{Spec}(\mathbb{Z})}$ by adding the archimedean place ∞. The arithmetic intersection theory on this compactified space involves both finite contributions (from primes) and archimedean contributions (from $\mathbb{R}$).

Our three-factor decomposition

$$\Omega_\Lambda = \underbrace{\frac{4}{3}}_{\text{global (gravity)}} \times \underbrace{2\pi}_{\text{archimedean}} \times \underbrace{\frac{1}{12}}_{\text{finite}}$$

mirrors the Arakelov product formula, where a global quantity equals a product of local contributions over all places.

Proposition 8.1 (Arakelov product interpretation)**.** *Define the local dark energy factors:*

$$\Omega_\Lambda^{(\infty)} = 2\pi \qquad \textit{(archimedean place)}, \tag{16}$$

$$\Omega_\Lambda^{(fin)} = \frac{1}{12} \qquad \textit{(finite places)}, \tag{17}$$

$$\Omega_\Lambda^{(grav)} = \frac{4}{3} \qquad \textit{(gravitational/global)}. \tag{18}$$

Then $\Omega_\Lambda = \Omega_\Lambda^{(grav)} \cdot \Omega_\Lambda^{(\infty)} \cdot \Omega_\Lambda^{(fin)}$*, which is analogous to the product formula* $\prod_v |x|_v = 1$ *in Arakelov theory.*

9 Predictions

The framework makes several testable predictions:

1. **No fourth force.** There is no additional gauge force beyond the Standard Model at any energy scale. Specifically, no SU(4) or higher gauge group appears as a fundamental force. This is testable at future colliders.

2. **Dark energy value.** $\Omega_\Lambda = 2\pi/9 \approx 0.6981$. The current Planck value is 0.6889 ± 0.0056; future surveys (Euclid, DESI, Rubin/LSST) will test this to sub-percent precision.

3. **No dark energy variation.** Since Ω_Λ is determined by the arithmetic constant $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$, the dark energy equation of state is exactly $w = -1$ (cosmological constant, not dynamical dark energy). This is testable via w_0–w_a measurements.

4. **Gauge coupling ratios.** The ratios of gauge couplings at some fundamental scale should reflect the dimensions $n = 1, 2, 3$ of the corresponding Galois representations. This prediction is less sharp and requires further development.

5. **Eisenstein/cusp ratio.** The ratio of vacuum energy to matter energy should be related to the ratio of Eisenstein to cuspidal contributions in the relevant space of modular forms.

10 Limitations

We are committed to intellectual honesty and list the principal limitations of the framework:

1. **The** $\mathrm{GL}(n) \leq \mathrm{cd}$ **bound is a heuristic, not a theorem.** The claim that Galois representations of dimension $n > \mathrm{cd}$ cannot "live" on $\mathrm{Spec}(\mathbb{Z})$ is motivated by the vanishing of cohomology in degrees above cd, but a rigorous proof would require a deeper understanding of the relationship between automorphic representations and étale cohomology in all dimensions. The Eichler–Shimura construction works for $n = 2$; generalizations to $n = 3$ and beyond are the subject of active research.

2. **Langlands for** $\mathrm{GL}(3)$ **is harder than** $\mathrm{GL}(2)$**.** While the Langlands correspondence for $\mathrm{GL}(2)/\mathbb{Q}$ is essentially established (Wiles, Taylor–Wiles, Breuil–Conrad–Diamond–Taylor), the correspondence for $\mathrm{GL}(3)/\mathbb{Q}$ is less complete. Attaching Galois representations to $\mathrm{GL}(3)$ automorphic forms has been achieved in many cases (e.g., by Clozel, Harris–Taylor, Scholze), but a complete picture remains work in progress.

3. **Conductor** $\to$ **mass does not work.** A natural hope is that the conductor (level) of a cusp form maps to the mass of the corresponding particle. This does not produce realistic particle masses, and we do not claim it does. The conductor–mass relationship, if it exists, must be more subtle than a direct identification.

4. $\mathrm{GL}(n)$ **vs.** $\mathrm{SU}(n)$**.** The passage from $\mathrm{GL}(n)$ (automorphic side) to $\mathrm{SU}(n)$ (gauge side) involves taking compact forms and modding out centers. This step is physically motivated but not derived from first principles within the arithmetic framework.

5. **The** 1.3% **discrepancy.** The predicted $\Omega_\Lambda = 2\pi/9 \approx 0.698$ is $\sim 1.3\%$ above the Planck central value 0.689. While within 2σ, this may indicate missing corrections (e.g., from higher-order arithmetic terms).

6. **No dynamics.** The framework determines the gauge group and spacetime dimension but does not derive the dynamics (Lagrangian) of the Standard Model. It is a "kinematic" framework, not a dynamical one.

11 Conclusion

The étale cohomological dimension $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$ is the "master number" of physics. From this single arithmetic invariant, we derive:

- Spacetime dimension: $d = \mathrm{cd} + 1 = 4$.
- Standard Model gauge group: $\mathrm{U}(1) \times \mathrm{SU}(2) \times \mathrm{SU}(3)$ from $\mathrm{GL}(1) \times \mathrm{GL}(2) \times \mathrm{GL}(3)$ with $n \leq \mathrm{cd} = 3$.
- Gauge boson count: $1 + 3 + 8 = 12 = 2/|B_2|$.
- Dark energy: $\Omega_\Lambda = (4/3) \times 2\pi \times (1/12) = 2\pi/9$.

The dark energy decomposes into three factors—gravitational, archimedean, and arithmetic—reflecting the Arakelov structure of $\mathrm{Spec}(\mathbb{Z})$. The archimedean factor (2π) explains why dark energy exists in our universe: because our universe is built on real numbers, not p-adic numbers.

The Eisenstein/cusp decomposition of modular forms provides a bridge between the vacuum/matter structure of physics and the arithmetic of $\mathrm{Spec}(\mathbb{Z})$, with Eisenstein series encoding the non-perturbative vacuum (left wing) and cusp forms encoding perturbative matter (right wing).

We believe that the path from $\mathrm{cd}(\mathrm{Spec}(\mathbb{Z})) = 3$ to the full structure of the Standard Model is one of the most promising directions in mathematical physics, bridging arithmetic geometry and fundamental physics in a way that neither field could achieve alone.

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